



PLC PRESBYTERIAN
LADIES' COLLEGE
SYDNEY
1888

2010
TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total Marks – 84

- Attempt questions 1-7
- All questions are of equal value

[illegible]

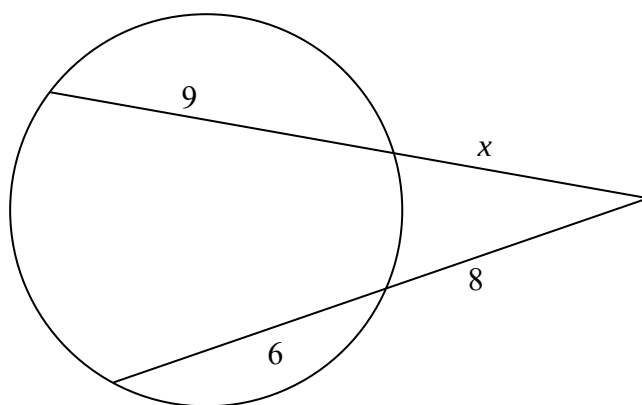
Question 1 (12 marks)**Start a new sheet of writing paper.****Marks**

- a) Show that $\lim_{x \rightarrow 0} \frac{3x}{\tan 2x} = \frac{3}{2}$. **1**
- b) $\int_0^{\frac{\pi}{8}} \sin^2 4x \, dx$ **3**
- c) Solve $\frac{4x-3}{x} \geq 5$ **3**
- d) Find the acute angle between the lines $y = -x - 1$ and $4x + 5y = 2$.
Answer to the nearest minute. **3**
- e) Using $t = \tan \frac{\theta}{2}$, find the exact value of $\frac{1 - \tan^2 15^\circ}{1 + \tan^2 15^\circ}$ showing all working. **2**

End of Question 1

Question 2 (12 marks)**Start a new sheet of writing paper.****Marks**

- a) Find the coordinates of the point, P , that divides the interval AB externally in the ratio of 1: 4 if $A(3, 1)$ and $B(-1, -5)$. **3**
- b) Using the substitution, $u = x^4 + 1$, or otherwise, evaluate $\int_0^1 x^3 e^{x^4+1} dx$. **3**
- c) Find the constant term in the expansion of $(x^2 - \frac{1}{2x^3})^{10}$ **2**
- d) Find the general solution to $2\cos\theta + \sqrt{3} = 0$. Express your answer in terms of π . **2**
- e) Find x , giving reasons: **2**

**End of Question 2**

Question 3 (12 marks)		Start a new sheet of writing paper.	Marks
a)	i)	Show that $x^3 + 2x - 17 = 0$ has a root between $x=2$ and $x=3$	1
	ii)	Using an approximation of $x = 2.4$, use one application of Newton's method to find a better approximation for this root. Give your answer to two decimal places.	3
b)	i)	Express $\sin x - 2\cos x$ in the form $A\sin(x - \alpha)$ where $0 \leq \alpha \leq \frac{\pi}{2}$.	2
	ii)	Hence, or otherwise, solve $\sin x - 2\cos x = \frac{\sqrt{5}}{2}$ for $0 \leq x \leq 2\pi$. Give your answer(s) correct to 2 decimal places.	2
c)		A particle moves on the x axis with velocity v m/s. The particle is initially at rest at $x=1$ m. Its acceleration is given by $\ddot{x} = 2v$ m/s ² . Find the velocity and acceleration of the particle at $x = 10$ metres.	4

End of Question 3

Question 4 (12 marks)**Start a new sheet of writing paper.****Marks**

a) Given $y = 2 \sin^{-1} \frac{x}{3}$

i) State the domain and range of this function.

2

ii) sketch the curve $y = 2 \sin^{-1} \frac{x}{3}$.

2

b) Find the exact value of $\int_0^{2\sqrt{3}} \frac{1}{4+x^2} dx$

2

c) Prove that $\frac{2}{\tan A + \cot A} = \sin 2A$

2

d) i) For the binomial expansion of $(4+3x)^{15}$, show that:

2

$$\frac{T_{k+1}}{T_k} = \frac{16-k}{k} \times \frac{3}{4} \times x.$$

ii) Hence, find the greatest coefficient of $(4+3x)^{15}$, leaving your answer in index form.

2**End of Question 4**

Question 5 (12 marks)**Start a new sheet of writing paper.****Marks**

- a) The velocity v m/s of a particle moving along the x -axis is given by $v^2 = 16x - 4x^2 + 20$.
- i) Prove that the motion is simple harmonic. **2**
- ii) Find the centre of motion. **1**
- iii) Find the distance travelled in one complete oscillation. **1**
- b) i) Show that the equation of the normal at $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ is $x + py - ap^3 - 2ap = 0$ **2**
- ii) The normals from $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ on the parabola $x^2 = 4ay$ meet at right angles. Prove that the locus of the points of intersection of these normals is the parabola $x^2 = ay - 3a^2$ **3**
- c) Prove by mathematical induction that $\frac{1}{2!} + \frac{2}{3!} + \dots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!}$ **3**
- for all positive integers n .

End of Question 5

Question 6 (12 marks)**Start a new sheet of writing paper.****Marks**

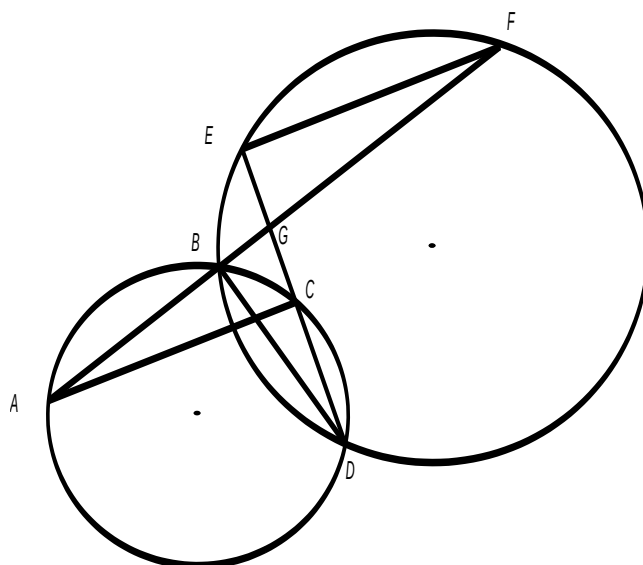
- a) Newton's law of cooling states that a body cools according to the equation $\frac{dT}{dt} = -k(T - S)$, where T is the temperature of the body at time t , S is the temperature of the surroundings and k is a constant.
- i) Show that $T = S + Ae^{-kt}$ satisfies the equation, where A is a constant. **1**
- ii) A metal rod has an initial temperature of 350°C and cools to 100°C in 10 minutes. The surrounding temperature is 24°C .
- (α) Find the value of A and show that $k = \frac{-1}{10} \log_e \left(\frac{38}{163} \right)$ **2**
- (β) Find how long it will take from the rod to cool to 25°C . **2**

Question 6 continues on the next page

Question 6 Continued

Marks

- b) In the diagram below, ABF and DCE are straight lines.

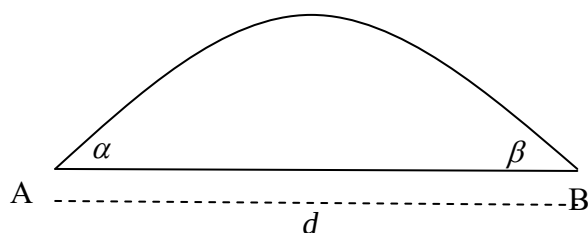


- i) Copy the diagram into your answer booklet.
 - ii) Prove that AC is parallel to EF . 3
- c) Given that $f(x) = \frac{(5-x)(1+x)}{5}$ and $h(x) = \log_e \{f(x)\}$
- i) Find the largest domain of $y = h(x)$. 1
 - ii) Find the equation of the inverse function $y = h^{-1}(x)$. 2
 - iii) Find the domain of the inverse function $y = h^{-1}(x)$. 1

End of Question 6

Question 7 (12 marks)**Start a new sheet of writing paper.****Marks**

- a) The polynomial $P(x) = x^3 - 2x^2 + ax + b$ has $(x+2)$ and $(x-2)$ as factors, find the
- i) values of a and b . 2
- ii) remaining root of $P(x) = x^3 - 2x^2 + ax + b$ 1
- b) A missile is launched from point A at an angle α and at a speed V towards a target at B , d metres away. Simultaneously a second missile is launched at speed W from B at an angle β , to intercept the first. The angles α and β are measured as in the diagram and are related by $\beta = 90^\circ - \alpha$.



The horizontal and vertical displacements of the projectiles from A and B are given by the following equations (DO NOT PROVE THESE RESULTS):

Missile from A :

$$x = Vt \cos \alpha$$

$$y = -\frac{1}{2}gt^2 + Vt \sin \alpha$$

Missile from B

$$x = d - Wt \cos \beta$$

$$y = -\frac{1}{2}gt^2 + Wt \sin \beta$$

- i) By equating y -components, show that if the missiles are to intersect, then the second missile must have speed $W = V \tan \alpha$. 2
- ii) Show that the time of intersection is $t = \frac{d \cos \alpha}{V}$ seconds after launch. 2

Question 7 continues on the next page

Question 7 Continued

Marks

c) Consider the function $f(x) = \frac{\log_e x}{x}$

i) Find the coordinates of the stationary point on the curve $y = f(x)$ and determine its nature. **3**

ii) Hence show that $\pi^e < e^\pi$ **2**

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Blank Page

Academic Year	Yr 12	Calendar Year	2010
Course	Ext. 1.	Name of task/exam	Ext. 1 TRIAL EXAM

Question 1:

$$\begin{aligned}
 \text{a) } \lim_{x \rightarrow 0} \frac{3x}{\tan 2x} &= \lim_{x \rightarrow 0} 3x \cdot \frac{x}{\tan 2x} \\
 &= \lim_{x \rightarrow 0} \frac{3x \cdot 2x}{2 \tan 2x} \\
 &= \lim_{x \rightarrow 0} \frac{3}{2} \left(\frac{2x}{\tan 2x} \right) \\
 &= \frac{3}{2} (1) \\
 &= \frac{3}{2}
 \end{aligned}$$

$$\text{b) } \int_0^{\frac{\pi}{8}} \sin^2 4x \, dx$$

$$\begin{aligned}
 \text{we know } \cos 8x &= 1 - 2 \sin^2 4x \\
 \therefore \sin^2 4x &= \frac{1}{2} - \frac{1}{2} \cos 8x
 \end{aligned}$$

$$\therefore \int_0^{\frac{\pi}{8}} \left(\frac{1}{2} - \frac{1}{2} \cos 8x \right) dx$$

$$= \left[\frac{1}{2} x - \frac{1}{16} \sin 8x \right]_0^{\frac{\pi}{8}}$$

$$= \left(\frac{1}{2} \left(\frac{\pi}{8} \right) - \frac{1}{16} \sin 8 \frac{\pi}{8} \right) - (0 - 0)$$

$$= \frac{\pi}{16}$$

$$\text{c) } \frac{4x-3}{x} \geq 5$$

$$x^2 \left(\frac{4x-3}{x} \right) \geq 5x^2$$

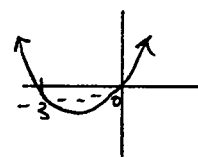
$$x(4x-3) - 5x^2 \geq 0$$

$$x[4x-3-5x] \geq 0$$

$$x(-3-x) \geq 0$$

$$x(x+3) \leq 0$$

$$\therefore -3 \leq x \leq 0$$



$$\begin{aligned}
 \text{d) } y &= -x - 1 & 4x + 5y &= 2 \\
 m &= -1 & m &= -\frac{4}{5}
 \end{aligned}$$

$$\therefore \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-1 - (-\frac{4}{5})}{1 + (-1)(-\frac{4}{5})} \right|$$

$$= \left| \frac{-\frac{1}{5}}{\frac{9}{5}} \right|$$

$$\tan \theta = \frac{1}{9}$$

$$\therefore \theta = 6^\circ 20'$$

$$\text{e) } t = \tan \frac{\theta}{2}$$

$$\therefore \cos \theta = \frac{1-t^2}{1+t^2}$$

$$\therefore \frac{1 - \tan^2 15^\circ}{1 + \tan^2 15^\circ} = \cos 30^\circ$$

$$\therefore \frac{1 - \tan^2 15^\circ}{1 + \tan^2 15^\circ} = \frac{\sqrt{3}}{2}$$

Academic Year	Yr 12	Calendar Year	2010
Course	Ext. 1.	Name of task/exam	TRIAL EXAM

Question 2:

$$a) A \begin{pmatrix} x_1 & y_1 \\ 3 & 1 \end{pmatrix} \quad B \begin{pmatrix} x_2 & y_2 \\ -1 & -5 \end{pmatrix} \quad -1:4$$

$$\therefore x = \frac{mx_2 + nx_1}{m+n}, \quad y = \frac{my_2 + ny_1}{m+n}$$

$$x = \frac{(-1)(-1) + 4(3)}{3}, \quad y = \frac{(-1)(-5) + 4(1)}{3}$$

$$x = \frac{1+12}{3}, \quad y = \frac{5+4}{3}$$

$$x = \frac{13}{3}, \quad y = 3$$

$$\therefore P\left(\frac{13}{3}, 3\right)$$

$$T_{k+1} = {}^{10}C_k x^{20-2k} \left(-\frac{1}{2}\right)^k x^{-3k}$$

$$= {}^{10}C_k \left(-\frac{1}{2}\right)^k x^{20-5k}$$

$$\text{for constant term } 20-5k=0$$

$$\therefore k=4$$

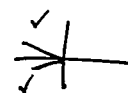
$$\therefore \text{constant term is } {}^{10}C_4 \left(-\frac{1}{2}\right)^4$$

$$= 210 \left(\frac{1}{16}\right)$$

$$= \frac{105}{8}$$

$$d) 2 \cos \theta + \sqrt{3} = 0$$

$$\cos \theta = -\frac{\sqrt{3}}{2}$$



$$\theta = (2k+1)\pi \pm \frac{\pi}{6}, \quad k \text{ integer}$$

$$b) \int_0^1 x^3 e^{x^4+1} dx$$

$$\therefore \int_0^1 \frac{1}{4} 4x^3 e^{x^4+1} dx \quad u = x^4 + 1$$

$$\frac{du}{dx} = 4x^3$$

$$du = 4x^3 dx$$

$$u=1 \text{ when } x=0$$

$$u=2 \text{ when } x=1$$

$$= \int_1^2 \frac{1}{4} e^u du$$

$$= \left[\frac{1}{4} e^u \right]_1^2$$

$$= \frac{1}{4} e^2 - \frac{1}{4} e^1$$

$$= \frac{1}{4} e (e-1)$$

$$e) (9+x)x = 14 \times 8 \quad (\text{The product of the intercepts of two intersecting secants to a circle from an external point are equal}).$$

$$9x + x^2 = 112$$

$$x^2 + 9x - 112 = 0$$

$$(x-7)(x+16) = 0$$

$$x = 7, -16$$

$$\text{but } x > 0 \text{ as length}$$

$$\therefore x = 7$$

$$c) \left(x^2 - \frac{1}{2x^3}\right)^{10}$$

$$T_{k+1} = {}^{10}C_k (x^2)^{10-k} \left(-\frac{1}{2x^3}\right)^k$$

Academic Year	Yr 12	Calendar Year	2010
Course	Ext. 1.	Name of task/exam	TRIAL EXAM

Question 3:

a) i) $P(x) = x^3 + 2x - 17$

$$P(2) = -5$$

$$P(3) = 16$$

Since $P(2) < 0$ and $P(3) > 0$
and $P(x)$ is continuous,
there exists a root between
 $x=2$ and $x=3$.

ii) $P(2.4) = 1.624$

$$P'(x) = 3x^2 + 2$$

$$P'(2.4) = 19.28$$

$$\therefore z_2 = z_1 - \frac{P(z_1)}{P'(z_1)}$$

$$= 2.4 - \frac{1.624}{19.28}$$

$$= 2.315 \dots$$

$\therefore$ A better root is $x = 2.32$
(2 dp)

b) $\sin x - 2 \cos x \equiv A \sin(x - \alpha)$

$$\sin x - 2 \cos x \equiv A \sin x \cos \alpha - A \cos x \sin \alpha$$

$$\therefore 1 = A \cos \alpha \quad (1)$$

$$2 = A \sin \alpha \quad (2)$$

$$1 + 4 = A^2 (\cos^2 \alpha + \sin^2 \alpha) \quad (1)^2 + (2)^2$$

$$5 = A^2$$

$$A = \sqrt{5}, \quad A > 0$$

$$\therefore 2 = \tan \alpha \quad (2) \div (1)$$

$$\alpha = 1.107 \dots$$

$$\therefore \sin x - 2 \cos x \equiv \sqrt{5} \sin(x - 1.107)$$

ii) $\sin x - 2 \cos x = \frac{\sqrt{5}}{2}$

$$\therefore \sqrt{5} \sin(x - 1.107) = \frac{\sqrt{5}}{2}$$

$$\sin(x - 1.107) = \frac{1}{2}$$

$$x - 1.107 = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\therefore x = 1.63, 3.73 \quad (2 \text{ dp})$$

c) $t=0 \quad v=0 \quad x=1 \text{ m.}$

$$\ddot{x} = 2v$$

$$v \frac{dv}{dx} = 2v$$

$$\frac{dv}{dx} = 2$$

$$\frac{dx}{dv} = \frac{1}{2}$$

$$x = \int \frac{1}{2} dv$$

$$x = \frac{1}{2} v + C$$

$$1 = 0 + C$$

$$\therefore x = \frac{1}{2} v + 1$$

at $x=10$

$$10 = \frac{1}{2} v + 1$$

$$9 = \frac{1}{2} v$$

$$v = 18 \text{ m/s}$$

$$\ddot{x} = 36 \text{ m/s}^2$$

Academic Year	Yr 12	Calendar Year	2010
Course	Ext. 1.	Name of task/exam	TRIAL EXAM

Question 4 :

a) $y = 2 \sin^{-1} \frac{x}{3}$

i) Domain : $-1 \leq \frac{x}{3} \leq 1$

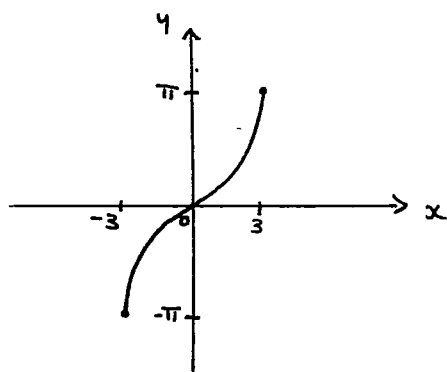
$\therefore -3 \leq x \leq 3$

Range : $-\frac{\pi}{2} \leq \sin^{-1} \frac{x}{3} \leq \frac{\pi}{2}$

$-\pi \leq 2 \sin^{-1} \frac{x}{3} \leq \pi$

$-\pi \leq y \leq \pi$

ii)



b) $\int_0^{2\sqrt{3}} \frac{1}{4+x^2} dx$

$= \frac{1}{2} \left[\tan^{-1} \frac{x}{2} \right]_0^{2\sqrt{3}}$

$= \frac{1}{2} \left[\tan^{-1} \frac{2\sqrt{3}}{2} - \tan^{-1} \frac{0}{2} \right]$

$= \frac{1}{2} \left[\tan^{-1} \sqrt{3} \right]$

$= \frac{1}{2} \cdot \frac{\pi}{3}$

$= \frac{\pi}{6}$

c) Prove $\frac{2}{\tan A + \cot A} = \sin 2A$

LHS = $\frac{2}{\frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}}$

$= \frac{2}{\frac{\sin^2 A + \cos^2 A}{\cos A \sin A}}$

$= \frac{2}{\frac{1}{\cos A \sin A}}$

$= 2 \cos A \sin A$

$= \sin 2A$

$= \text{RHS}$

$\therefore \text{proved}$

d) i) $(4+3x)^{15}$

$T_{k+1} = {}^{15}C_k (4)^{15-k} (3x)^k$

$T_k = {}^{15}C_{k-1} (4)^{15-(k-1)} (3x)^{k-1}$

$\therefore \frac{T_{k+1}}{T_k} = \frac{\frac{15!}{k!(15-k)!} 4^{15-k} 3^k x^k}{\frac{15!}{(k-1)!(16-k)!} 4^{16-k} 3^{k-1} x^{k-1}}$

$= \frac{15!}{k!(15-k)!} \times \frac{(k-1)!(16-k)!}{15!} \times 4^{15-k-(16-k)} \times \frac{3^k}{3^{k-1}} \times \frac{x^k}{x^{k-1}}$

$= \frac{(16-k)}{k} \times 4^{-1} \times 3 \times x$

$\frac{T_{k+1}}{T_k} = \frac{16-k}{k} \times \frac{3}{4} x$

Academic Year	Yr 12	Calendar Year	2010
Course	Ext. 1.	Name of task/exam	TRIAL EXAM

ii) for greatest coefficient

$$\frac{T_{k+1}}{T_k} > 1$$

$$\frac{16-k}{k} \times \frac{3}{4} > 1$$

$$48 - 3k > 4k \quad (\text{as } k > 0)$$

$$48 > 7k$$

$$k < \frac{48}{7}$$

$$\therefore k = 6, 5, \dots$$

$$\therefore k = 6 \text{ is greatest}$$

$$\begin{aligned} \therefore T_7 &= {}^{15}C_6 (4)^{15-6} (3)^6 \\ &= 5005 \times 4^9 \times 3^6 \\ &= 5005 \times 2^{18} \times 3^6 \end{aligned}$$

Question 5:

a) $v^2 = 16x - 4x^2 + 20$

i) $\frac{1}{2}v^2 = 8x - 2x^2 + 10$

$$\frac{d}{dx} \left(\frac{1}{2}v^2 \right) = \frac{d}{dx} (8x - 2x^2 + 10)$$

$$\ddot{x} = 8 - 4x$$

$$\ddot{x} = -4(x - 2)$$

which is of the form

$$\ddot{x} = -n^2(x - h) \text{ which is}$$

s.h.m with centre at $x = h$

$$\text{i.e. } x = 2.$$

ii) Centre of motion is $x = 2$

iii) when $v = 0$

$$0 = 16x - 4x^2 + 20$$

$$4x^2 - 16x - 20 = 0$$

$$x^2 - 4x - 5 = 0$$

$$(x+1)(x-5) = 0$$

oscillates between -1 & 5

$\therefore$ distance travelled is 12 m.

b) i) $x^2 = 4ay$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

$$\text{at } x = 2ap$$

$$\begin{aligned} m_{\text{tang}} &= \frac{4ap}{4a} \\ &= p. \end{aligned}$$

$$\therefore m_{\text{normal}} = -\frac{1}{p}$$

$\therefore$ equation normal:

$$y - y_1 = m(x - x_1)$$

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$\therefore x + py - ap^3 - 2ap = 0$$

ii) If normals meet at right angles then

$$-\frac{1}{p} \times -\frac{1}{q} = -1$$

$$\therefore pq = -1$$

Academic Year	Yr 12	Calendar Year	2010
Course	Ext. 1.	Name of task/exam	TRIAL EXAM

eqn of normals

$$x + py - ap^3 - 2ap = 0 \quad (1)$$

$$x + qy - aq^3 - 2aq = 0 \quad (2)$$

$$y(p-q) - ap^3 + aq^3 - 2ap + 2aq = 0 \quad (1) - (2)$$

$$y(p-q) - a(p^3 - q^3) - 2a(p-q) = 0$$

$$y(p-q) - a(p-q)(p^2 + pq + q^2) - 2a(p-q) = 0$$

since $p \neq q$ can divide by $(p-q)$

$$y - a(p^2 + pq + q^2) - 2a = 0$$

$$y = a(p^2 + pq + q^2) + 2a$$

$$x = -py + ap^3 + 2ap$$

$$= -p[a(p^2 + pq + q^2) + 2a] + ap^3 + 2ap$$

$$= -ap^3 - p^2qa - apq^2 - 2ap + ap^3 + 2ap$$

$$x = -apq(p+q)$$

 $\therefore$ pt intersection of Normals:

$$(-apq(p+q), a(p^2 + pq + q^2 + 2))$$

Locus:

we know $pq = -1$ and

$$x = -apq(p+q) \quad y = a(p^2 + pq + q^2 + 2)$$

$$x = -a(-1)(p+q)$$

$$x = a(p+q)$$

$$\frac{x}{a} = p+q$$

$$y = a(p^2 + pq + q^2) + 2a$$

$$\frac{y-2a}{a} = p^2 + (-1) + q^2$$

$$\frac{y-2a}{a} + 1 = p^2 + q^2$$

$$\frac{y-2a+a}{a} = (p+q)^2 - 2pq$$

$$\frac{y-a}{a} = \left(\frac{x}{a}\right)^2 - 2(-1)$$

$$\frac{y-a}{a} = \frac{x^2}{a^2} + 2$$

$$a(y-a) = x^2 + 2a^2$$

$$\therefore x^2 = ay - a^2 - 2a^2$$

$$x^2 = ay - 3a^2$$

c) Prove

$$\frac{1}{2!} + \frac{2}{3!} + \dots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!}$$

for all positive integers n .Step 1: prove true for $n=1$

$$\begin{aligned} \text{LHS} &= \frac{1}{2!} \\ &= \frac{1}{2} \end{aligned}$$

$$\text{RHS} = \frac{(1+1)! - 1}{(1+1)!}$$

$$= \frac{2! - 1}{2!}$$

$$= \frac{2-1}{2}$$

$$= \frac{1}{2}$$

$$\therefore \text{LHS} = \text{RHS}$$

 $\therefore$ true for $n=1$

Academic Year	Yr 12	Calendar Year	2010
Course	Ext. 1.	Name of task/exam	TRIAL EXAM

Step 2 : Assume true for $n=k$

$$\therefore \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} = \frac{(k+1)! - 1}{(k+1)!}$$

Step 3 : Prove true for $n=k+1$

i.e. prove,

$$\frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k+1}{(k+2)!} = \frac{(k+2)! - 1}{(k+2)!}$$

$$\text{LHS} = \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= \frac{(k+1)! - 1}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= \frac{(k+2)[(k+1)! - 1] + k+1}{(k+2)!}$$

$$= \frac{(k+2)! - (k+2) + k+1}{(k+2)!}$$

$$= \frac{(k+2)! - \cancel{k-2} + \cancel{k+1}}{(k+2)!}$$

$$= \frac{(k+2)! - 1}{(k+2)!}$$

= RHS

$\therefore$ by the principle of mathematical induction, proved for all positive integers, n .

Question 6 :

a) $\frac{dT}{dt} = -k(T-S)$

i) $T = S + Ae^{-kt}$

$$T - S = Ae^{-kt} \dots\dots\dots \textcircled{1}$$

also

$$\frac{dT}{dt} = Ae^{-kt} (-k)$$

$$= (T-S) (-k) \text{ from } \textcircled{1}$$

$$\frac{dT}{dt} = -k(T-S)$$

ii) $t=0 \quad T=350^\circ\text{C}$

$t=10 \text{ min} \quad T=100^\circ\text{C}$

$S=24^\circ\text{C}$

(*) $T = 24 + Ae^{-kt}$

$$350 = 24 + Ae^0$$

$$A = 326$$

$$\therefore T = 24 + 326e^{-kt}$$

$$100 = 24 + 326e^{-k \times 10}$$

$$76 = 326e^{-10k}$$

$$\frac{76}{326} = e^{-10k}$$

$$\ln\left(\frac{76}{326}\right) = -10k$$

$$k = -\frac{1}{10} \ln\left(\frac{76}{326}\right)$$

$$k = -\frac{1}{10} \ln\left(\frac{38}{163}\right)$$

Academic Year	Yr 12	Calendar Year	2010
Course	Ext. 1.	Name of task/exam	TRIAL EXAM

$$(f) \quad T = 24 + 326e^{-kt}$$

$$25 = 24 + 326e^{-kt}$$

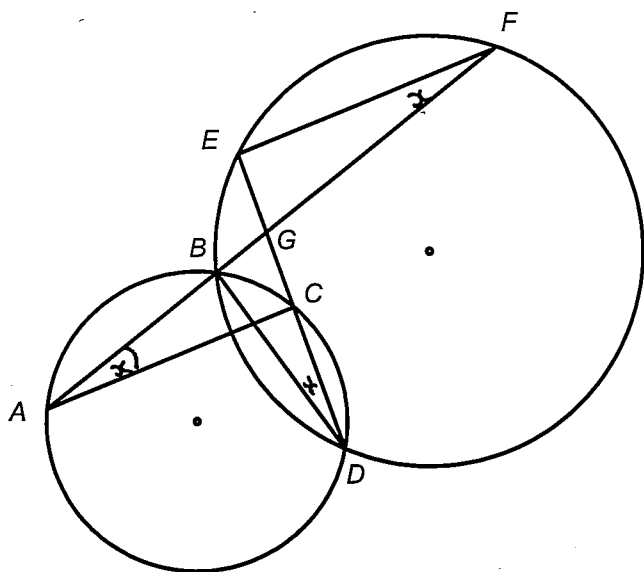
$$\frac{1}{326} = e^{-kt}$$

$$\ln\left(\frac{1}{326}\right) = -kt$$

$$t = -\frac{1}{k} \ln\left(\frac{1}{326}\right)$$

$$t = 39.7 \text{ mins}$$

b) i)



ii) Let $\angle BAC = x$

$\therefore \angle BDC = x$ (angles standing on same arc BC are equal)

$\therefore \angle BFE = x$ (angles standing on same arc BE equal)

$\therefore \angle BAC = \angle EFG$ and they are alternate angles.

$\therefore AC \parallel EF$

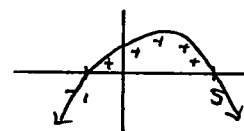
$$c) \quad f(x) = \frac{(5-x)(1+x)}{5}$$

$$h(x) = \ln\{f(x)\}$$

$$i) \quad D: f(x) > 0$$

$$\therefore \frac{(5-x)(1+x)}{5} > 0$$

$$(5-x)(1+x) > 0$$



$$\therefore -1 < x < 5$$

$$ii) \quad f: y = \ln\left[\frac{(5-x)(1+x)}{5}\right]$$

$$f^{-1}: x = \ln\left[\frac{(5-y)(1+y)}{5}\right]$$

$$e^x = \frac{(5-y)(1+y)}{5}$$

$$5e^x = (5-y)(1+y)$$

$$5e^x = 5 + 4y - y^2$$

$$-5e^x = y^2 - 4y - 5$$

$$-5e^x = y^2 - 4y + 4 - 9$$

$$9 - 5e^x = (y-2)^2$$

$$y-2 = \pm\sqrt{9-5e^x}$$

$$y = 2 \pm \sqrt{9-5e^x}$$

for function, either of these both not both.

Academic Year	Yr 12	Calendar Year	2010
Course	Ext. 1	Name of task/exam	TRIAL EXAM

iii) Domain: $9 - 5e^x \geq 0$
 $5e^x \leq 9$
 $e^x \leq \frac{9}{5}$
 $x \ln e \leq \ln \frac{9}{5}$
 $x \leq \ln \frac{9}{5}$

Question 7:

a) i) $P(x) = x^3 - 2x^2 + ax + b$

$P(-2) = 0$ and $P(2) = 0$

$\therefore -8 - 8 - 2a + b = 0$ ①

$8 - 8 + 2a + b = 0$ ②

$-2a + b = 16$ ①

$2a + b = 0$ ②

$2b = 16$ ① + ②

$b = 8$

$a = -4$

ii) $\therefore P(x) = x^3 - 2x^2 - 4x + 8$
 $= (x+2)(x-2)(x-\alpha)$

$$\begin{array}{r} x^2 - 4 \overline{) x^3 - 2x^2 - 4x + 8} \\ \underline{x^3 - 4x} \\ -2x^2 + 8 \\ \underline{-2x^2 + 8} \\ 0 \end{array}$$

$\therefore$ other root is $x = 2$.

$\therefore$ roots are $2, 2, -2$

b) i) $y = -\frac{1}{2}gt^2 + vt \sin \alpha$ ①

$y = -\frac{1}{2}gt^2 + wt \sin \beta$

~~$-\frac{1}{2}gt^2 + vt \sin \alpha = -\frac{1}{2}gt^2 + wt \sin \beta$~~

$vt \sin \alpha = wt \sin \beta$

$v \sin \alpha = w \sin \beta$

$\beta = 90 - \alpha$

$\therefore v \sin \alpha = w \sin (90 - \alpha)$

$v \sin \alpha = w \cos \alpha$

$\therefore w = \frac{v \sin \alpha}{\cos \alpha}$

$w = v \tan \alpha$

ii) $Vt \cos \alpha = d - Wt \cos \beta$

$Vt \cos \alpha + Wt \cos \beta = d$

$t(V \cos \alpha + W \cos \beta) = d$

$t = \frac{d}{(V \cos \alpha + v \tan \alpha \cos \beta)}$

$= \frac{d}{(V \cos \alpha + v \tan \alpha \cos (90 - \alpha))}$

$= \frac{d}{(V \cos \alpha + v \frac{\sin \alpha}{\cos \alpha} \cdot \sin \alpha)}$

$t = \frac{d}{\frac{V \cos^2 \alpha + v \sin^2 \alpha}{\cos \alpha}}$

Academic Year	Yr 12	Calendar Year	2010
Course	Ext. 1.	Name of task/exam	Trial Exam

$$t = \frac{d \cos \alpha}{v(\cos^2 \alpha + \sin^2 \alpha)}$$

$$t = \frac{d \cos \alpha}{v}$$

$$c) f(x) = \frac{\ln x}{x}$$

$$\begin{aligned} i) f'(x) &= \frac{x \left(\frac{1}{x} \right) - \ln x (1)}{x^2} \\ &= \frac{1 - \ln x}{x^2} \end{aligned}$$

$$\text{for stat. pts } f'(x) = 0$$

$$\therefore \frac{1 - \ln x}{x^2} = 0$$

$$1 - \ln x = 0$$

$$\ln x = 1$$

$$\therefore e = x$$

$$\begin{aligned} \text{when } x = e \quad y &= \frac{\ln e}{e} \\ &= \frac{1}{e} \end{aligned}$$

$$\therefore \text{Stat. pt } \left(e, \frac{1}{e} \right)$$

$$\begin{aligned} f''(x) &= \frac{x^2 \left(-\frac{1}{x} \right) - (1 - \ln x) 2x}{x^4} \\ &= \frac{-x - 2x + 2x \ln x}{x^4} \end{aligned}$$

$$\therefore f''(x) = -\frac{3x + 2x \ln x}{x^4}$$

$$\text{when } x = e$$

$$f''(x) = -\frac{3e + 2e \ln e}{e^4}$$

$$= -\frac{3e + 2e}{e^4}$$

$$= -\frac{1}{e^3}$$

$$< 0$$

$$\therefore \text{max at } \left(e, \frac{1}{e} \right).$$

$$ii) \text{ Since max value is } \frac{1}{e}$$

$$\text{then } \frac{\ln x}{x} < \frac{1}{e}$$

$$\text{at } x = \pi$$

$$\frac{\ln \pi}{\pi} < \frac{1}{e}$$

$$e \ln \pi < \pi$$

$$\ln \pi^e < \pi$$

take e of both sides

$$e^{\ln \pi^e} < e^\pi$$

$$\therefore \pi^e < e^\pi$$